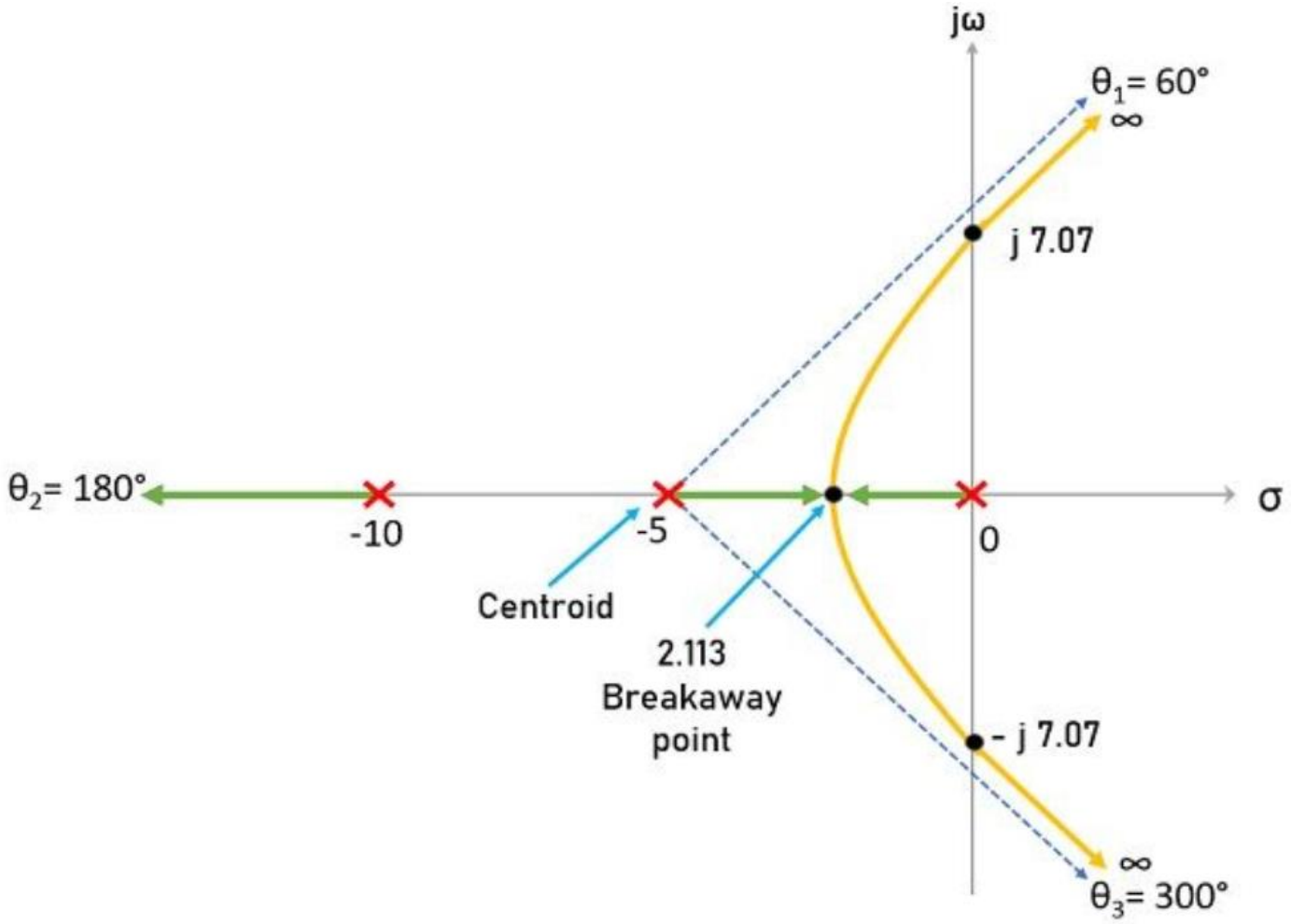
# Root-Locus

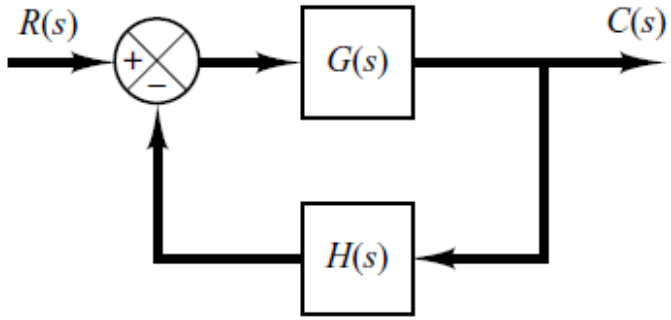
Understanding the importance of poles in a feedback control



# Objectives

- MLO #1.1: Formulate and interpret a root locus to determine the effects of a parameter variation (usually a positive  gain in negative feedback) on the closed loop poles of a system.
- MLO #1.2: Based on the properties of root loci (loci is the plural of locus, but I don’t like it), manually construct them
- MLO #1.3: Use Matlab to construct root loci and interpret the closed loop response.

# Simple example




$$G(s) = \frac{K}{s(s+1)} \qquad H(s) = 1$$

$$\frac{C(s)}{R(s)} = \frac{1}{1+GH} = \frac{1}{s^2+s+K}$$

Closed Loop poles as a function of K (K: 0→∞):

$$s_1 = -\frac{1}{2} + \frac{1}{2}\sqrt{1-4K}$$

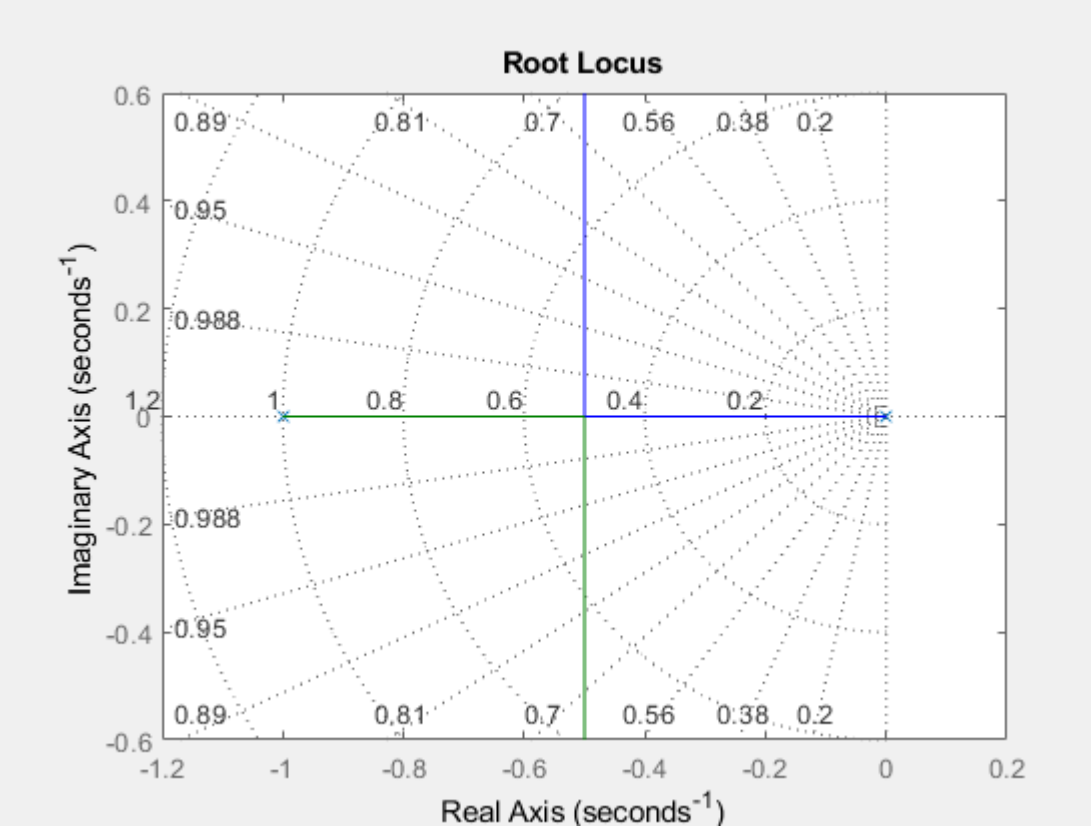
$$s_1 = -\frac{1}{2} - \frac{1}{2}\sqrt{1-4K}$$

Notice that the location of the two poles $s_1$ and $s_2$ are functions of $K$. The root locus consists of plotting $s_1$ and $s_2$ in the s-plane as K varies from 0 to ∞.

# Root Locus of Simple Example

```
>> s = tf('s');
>> G = 1/(s*(s+1));
>> rlocus(G)
```

Use MATLAB to find the pole locations for $K = 1, K = 5$ and $K = 10$. Verify with the locations given by the analytical equations from the previous slides.



# Root Locus Conditions

- We are interested in the roots of the characteristic equation:

$$1 + G(s)H(s) = 0$$
$$G(s)H(s) = -1$$

- Given that $s$ is complex variable ($s = \sigma + j\omega$), the last equation above $G(s)H(s) = -1$ is indeed an equality between two complex numbers. If $s$ is a point in the complex plane and if $s$ belongs to the root locus, then we state the two root-locus conditions:
- Angle Condition:

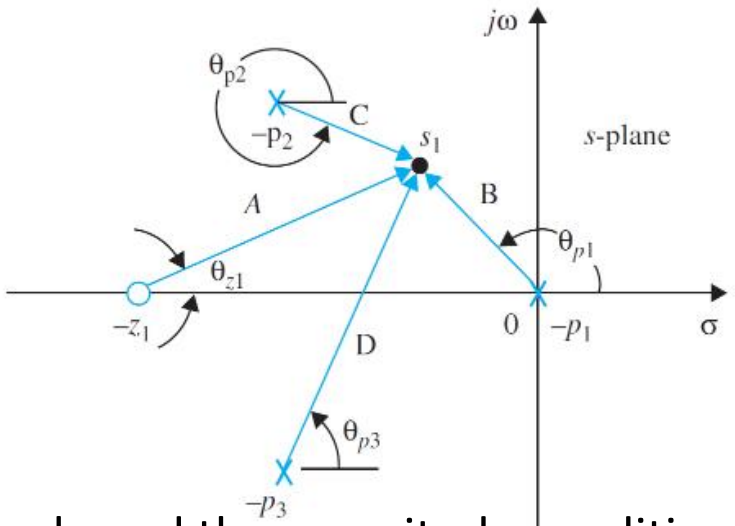
$$\angle G(s)H(s) = \angle -1 = \pm 180°(2\ell + 1) \quad (\ell = 0, 1, 2, \dots)$$

- Magnitude Condition:

$$|G(s)H(s)| = 1$$

# General Example

$$G(s)H(s)=\frac{K(s+z_1)}{s(s+p_2)(s+p_3)}$$




For $s_1$ to belong to the root locus, $s_1$ must satisfy the angle and the magnitude condition; namely,

1) angle condition must be satisfied first:

$$\angle(s_1+z_1)-\angle s_1-\angle(s_1+p_2)-\angle(s_1+p_3)=$$
$$\theta_{z1}-\theta_{p1}-\theta_{p2}-\theta_{p3}=\pm 180°(2\ell+1)$$

2) Magnitude condition:

$$\frac{|K||s_1+z_1|}{|s_1||s_2+p_2||s_3+p_3|}=1$$

Once the angle condition is satisfied, the magnitude condition can be forced as follows:

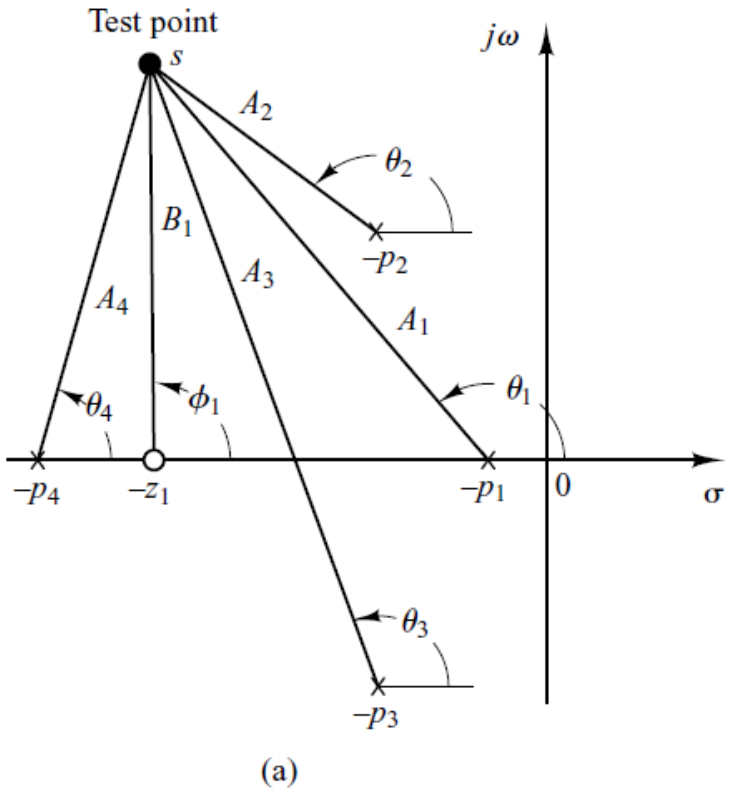
$$|K|=\frac{|s_1||s_2+p_2||s_3+p_3|}{|s_1+z_1|}$$

# Another Example

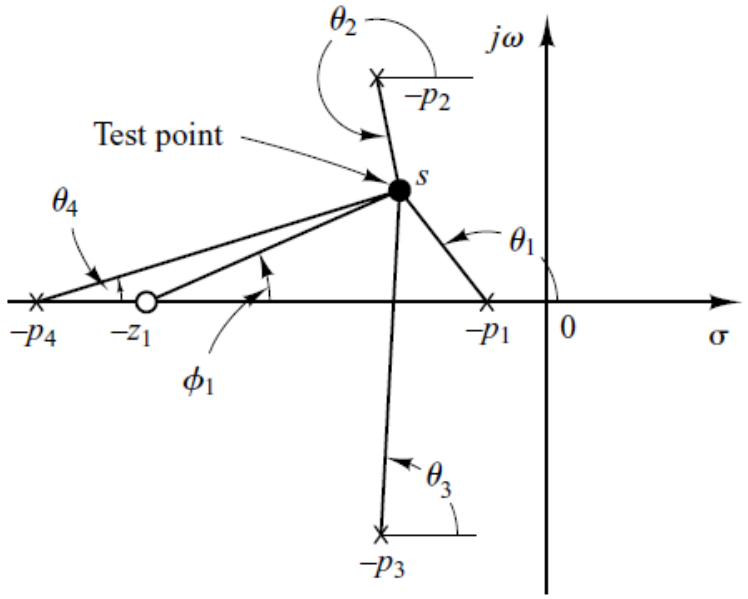
$$G(s)H(s) = \frac{K(s + z_1)}{(s + p_1)(s + p_2)(s + p_3)(s + p_4)}$$

$$\underline{/G(s)H(s)} = \phi_1 - \theta_1 - \theta_2 - \theta_3 - \theta_4$$

$$|G(s)H(s)| = \frac{KB_1}{A_1A_2A_3A_4}$$



(a)



(b)

# Properties and Rules to Construct a Root Locus Manually (1)

- 1) Locate the open-loop zeros and poles in the s complex plane. The root locus starts (K=0) at the poles of the open loop transfer function and ends ($K \rightarrow \infty$) at the zeros of the open loop transfer function.
- 2) Determine the number of branches. The number of branches equal the number of poles of the open loop transfer function (that is, the order of the denominator polynomial of the open loop transfer function). A branch in a RL is the locus of one root when K varies from zero to infinity.

# Properties and Rules to Construct a Root Locus Manually (2)

- 3) Determine the root loci on the real axis. Root loci on the real axis are determined by open-loop poles and zeros lying on it. Choose a test point on the real axis. If the number of real (open loop) zeros and poles on the right of such test point is an odd number, then that test point belongs to the root locus.
- 4) Determine the asymptotes of root loci. Asymptotes of root loci refer to the behavior of root loci when $|s| \to \infty$. There are $n - m$ asymptotes, where $n$ is the order of the denominator of $GH$ and $m$ is the order of the denominator of $GH$.

# Properties and Rules to Construct a Root Locus Manually (3)

- 5) Determine the angle of the asymptotes (only if $n \neq m$)

$$\text{Angle of asymptotes} = \frac{\pm 180°(2\ell + 1)}{n - m} \qquad (\ell = 0,1,2, \dots)$$

$$n = \text{number of finite poles of } G(s)H(s)$$
$$m = \text{number of finite zeros of } G(s)H(s)$$

- The RL is symmetrical with respect to the real axis. Why? Because complex poles always come in pair of complex conjugates. Make sure any branch or asymptote in the upper complex plane has its symmetrical branch or asymptote in the lower complex plane.

# Properties and Rules to Construct a Root Locus Manually (4)

- 6) Determine the point of intersection of all asymptotes:

$$\sigma_1 = \frac{\sum \text{finite poles of } G(s)H(s) - \sum \text{finite zeros of } G(s)H(s)}{n-m} \qquad (9\text{-}35)$$

- 7) Find the breakaway and break-in points. See next slides to understand the procedure of finding the breakaway points. This requires the calculation of some derivatives.

# Properties and Rules to Construct a Root Locus Manually (5)

- “If a root locus lies between two adjacent open-loop poles on the real axis, then there exists at least one breakaway point between the two poles. Similarly, if the root locus lies between two adjacent zeros (one zero may be located at $-\infty$) on the real axis, then there always exists at least one break-in point between the two zeros. If the root locus lies between an open-loop pole and a zero (finite or infinite) on the real axis, then there may exist no breakaway or break-in points or there may exist both breakaway and break-in points.”
- Rewrite $G(s)H(s)$ as:

$$G(s)H(s) = \frac{KA(s)}{B(s)}$$

$$G(s)H(s) + 1 = \frac{B(s) + KA(S)}{B(s)} = 0 \Rightarrow B(s) + KA(s) = 0 \quad (1)$$

# Properties and Rules to Construct a Root Locus Manually (6)

- From equation (1):

$$K(s) = -\frac{B(s)}{A(s)}$$

- Differentiate $K(s)$ with respect to s:

$$\frac{dK}{ds} = -\frac{B'(s)A(s) - B(s)A'(s)}{A^2(s)} = 0$$

- Finally, the breakaway points are found by solving the algebraic equation:

$$B'(s)A(s) - B(s)A'(s) = 0$$

# Properties and Rules to Construct a Root Locus Manually

- 8) Determine the angle of departure (angle of arrival) of the root locus from a complex pole (at a complex zero)

Angle of departure from a complex pole $= 180^\circ$
$-$ (sum of the angles of vectors to a complex pole in question from other poles)
$+$ (sum of the angles of vectors to a complex pole in question from zeros)

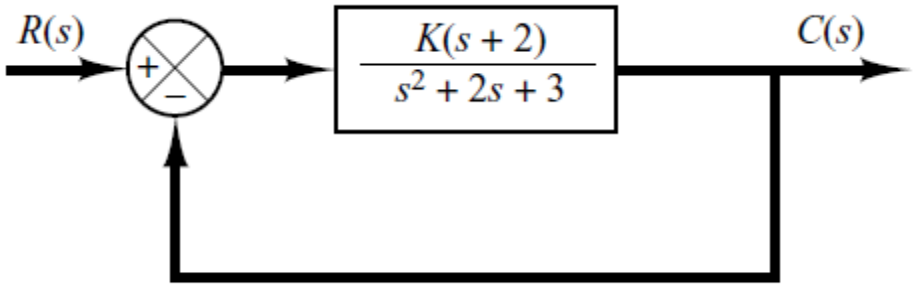
Angle of arrival at a complex zero $= 180^\circ$
$-$ (sum of the angles of vectors to a complex zero in question from other zeros)
$+$ (sum of the angles of vectors to a complex zero in question from poles)

# Properties and Rules to Construct a Root Locus Manually

- 9) Find the points where the root loci may cross the imaginary axis. Routh's stability criterion applied to the characteristic equation is a good tool for this, however less laborious methods can be used for this.

# Example

- Sketch the root locus of the following system



# Example

- Sketch the root locus of the following system

